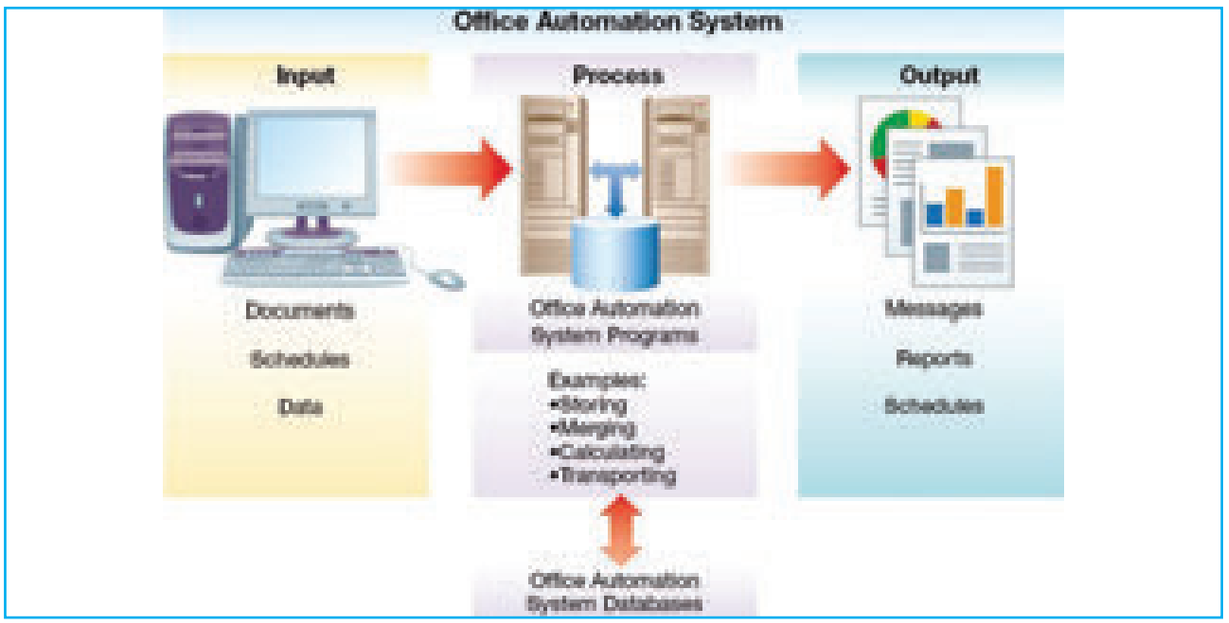
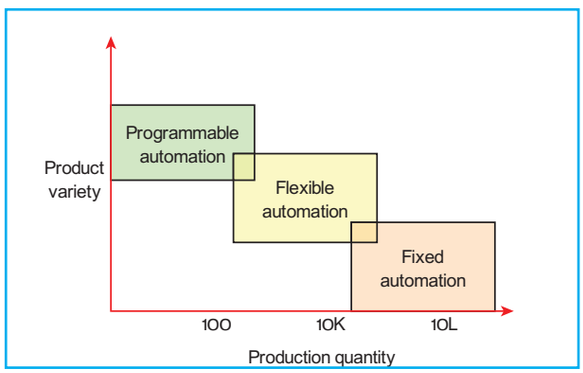


Office Automation

A. Fixed Automation (Hard Automation)

Fixed automation is a system in which the sequence of processing (or assembly) operations is **fixed** by the equipment configuration.



Advantages

1. The movement of parts is very fast and efficient
2. Very high efficiency
3. Unit cost is low

Disadvantages

1. High initial investment
2. Inflexibility

B. Programmable Automation

The word "Programmable" means that one set of task can be easily switched over to another set by changing the computerized instructions.

i. Computer Aided Design (CAD)

ii. Computer aided Manufacturing (CAM)

iii. Computer Integrated Manufacturing (CIM)

Advantages

- For large batches the unit cost is low
- The change in product can be dealt with flexibility

Disadvantages

- Unit cost, as compared to fixed automation is high
- Long setup time is required for new product

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